

Assignment 2 Symbolic Calculation

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Problem 1

$$\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{x^3}}.$$

No limit.

```
1  %-----Problem 1-----
2  clc; clear; close all;
3
4  syms x;
5  result = limit((cos(x))^(x^(-3)), x, 0);
6
7  if isnan(result)
8      disp('Problem 1: No limit.');
```

```
9  else
10     fprintf('The answer of problem 1 is: %s\n', char(result));
11 end
12
13 % Output:
14 % Problem 1: No limit.
```

Problem 2

$$\ln x + e^{\frac{y^2}{x}} = e.$$
$$\frac{dy}{dx} = -\frac{e^{-y^2/x}(1 - \frac{y^2}{x}e^{y^2/x})}{2y}$$

```

1  %-----Problem 2-----
2  clc; clear; close all;
3
4  syms x y;
5  F = log(x) + exp(y^2/x) - exp(1);
6  % Implicit Differentiation Rule
7  dFdx = diff(F, x);
8  dFdy = diff(F, y);
9  dydx = -dFdx / dFdy;
10
11 if isnan(dydx) || isinf(dydx)
12     disp('Problem 2: No derivative. ');
13 else
14     fprintf('The answer of problem 2 is: ');
15     disp(dydx);
16 end
17
18 % Output:
19 % The answer of problem 2 is: -(x*exp(-y^2/x)*(1/x - ...
    (y^2*exp(y^2/x))/x^2))/(2*y)

```

Problem 3

$$\begin{cases} x = a(t^3 - \sin^2 t) \\ y = a(t - \cos t) \end{cases}$$

$$\frac{dy}{dx} = \frac{-(\sin t + 1)}{2 \cos t \sin t - 3t^2}$$

```

1  %-----Problem 3-----
2  clc; clear; close all;
3
4  syms a t;
5  x = a * (t^3 - (sin(t))^2);
6  y = a * (t - cos(t));
7  % Differentiate Parametric Equations
8  dxdt = diff(x, t);
9  dydt = diff(y, t);
10 dydx = dydt / dxdt;
11
12 if isinf(dydx) || isnan(dydx)
13     disp('Problem 3: No derivative. ');
14 else
15     fprintf('The answer of problem 3 is: ');
16     disp(dydx);
17 end
18
19 % Output:

```

```

20 % The answer of problem 3 is:  $-(\sin(t) + 1)/(2*\cos(t)*\sin(t) - ...$ 
     $3*t^2)$ 

```

Problem 4

$$\sin(x + y) + \cos(y^2 + z^2) + \tan(x + z^3) = 0.$$

$$\frac{\partial z}{\partial x} = -\frac{\cos(x + y) + \tan((z^3 + x)^2) + 1}{3z^2(\tan((z^3 + x)^2) + 1) - 2z \sin(y^2 + z^2)}$$

$$\frac{\partial z}{\partial y} = -\frac{\cos(x + y) - 2y \sin(y^2 + z^2)}{3z^2(\tan((z^3 + x)^2) + 1) - 2z \sin(y^2 + z^2)}$$

```

1  %-----Problem 4-----
2  clc; clear; close all;
3
4  syms x y z;
5  F = sin(x + y) + cos(y^2 + z^2) + tan(x + z^3);
6  % Implicit Differentiation Rule
7  dFdx = diff(F, x);
8  dFdy = diff(F, y);
9  dFdz = diff(F, z);
10 dpzdpdx = -dFdx / dFdz;
11 dpzdpdy = -dFdy / dFdz;
12
13 if isnan(dpzdpdx) || isinf(dpzdpdx)
14     disp('Problem 4(a): No partial derivative. ');
15 else
16     fprintf('The answer of problem 4(a) is: ');
17     disp(dpzdpdx);
18 end
19
20 if isnan(dpzdpdy) || isinf(dpzdpdy)
21     disp('Problem 4(b): No partial derivative. ');
22 else
23     fprintf('The answer of problem 4(b) is: ');
24     disp(dpzdpdy);
25 end
26
27 % Output:
28 % The answer of problem 4(a) is:  $-(\cos(x + y) + \tan(z^3 + x)^2 + ...$ 
     $1)/(3*z^2*(\tan(z^3 + x)^2 + 1) - 2*z*\sin(y^2 + z^2))$ 
29
30 % The answer of problem 4(b) is:  $-(\cos(x + y) - 2*y*\sin(y^2 + ...$ 
     $z^2))/(3*z^2*(\tan(z^3 + x)^2 + 1) - 2*z*\sin(y^2 + z^2))$ 

```

Problem 5

$$u(x, y, z) = \cos(x + y)e^{x^2 y \sin z}.$$

$$\left. \frac{\partial^5 u(x, y, z)}{\partial x^2 \partial y^2 \partial z} \right|_{(1,2,1)} = -334.1558$$

```

1  %-----Problem 5-----
2  clc; clear; close all;
3
4  syms x y z;
5  u = cos(x + y) * exp(x^2 * y * sin(z));
6
7  dudx2 = diff(u, x, 2);
8  dudy2 = diff(dudx2, y, 2);
9  dudz = diff(dudy2, z, 1);
10
11 if isnan(dudz) || isinf(dudz)
12     disp('Problem 5: No partial derivative.');
```

The answer of problem 5 is:

```

13 else
14     result = subs(dudz, [x, y, z], [1, 2, 1]);
15     fprintf('The answer of problem 5 is: ');
16     disp(result);
17 end
18
19 % Output:
20 % The answer of problem 5 is: 6*cos(1)*exp(2*sin(1))*sin(3) - ...
    10*cos(1)*cos(3)*exp(2*sin(1)) + ...
    66*cos(1)*cos(3)*exp(2*sin(1))*sin(1)^2 + ...
    136*cos(1)*cos(3)*exp(2*sin(1))*sin(1)^3 + ...
    32*cos(1)*cos(3)*exp(2*sin(1))*sin(1)^4 - ...
    216*cos(1)*exp(2*sin(1))*sin(1)^2*sin(3) - ...
    80*cos(1)*exp(2*sin(1))*sin(1)^3*sin(3) - ...
    66*cos(1)*cos(3)*exp(2*sin(1))*sin(1) - ...
    84*cos(1)*exp(2*sin(1))*sin(1)*sin(3)
```

Problem 6

$$\int \sin^{10} x dx = \frac{63x}{256} - \frac{105 \sin(2x)}{512} + \frac{15 \sin(4x)}{256} - \frac{15 \sin(6x)}{1024} + \frac{5 \sin(8x)}{2048} - \frac{\sin(10x)}{5120}$$

```

1  %-----Problem 6-----
2  clc; clear; close all;
3
4  syms x;
5  result = int((sin(x))^10, x);
```

```

6
7 fprintf('The answer of problem 6 is: ');
8 disp(result);
9
10 % Output:
11 % The answer of problem 6 is: (63*x)/256 - (105*sin(2*x))/512 + ...
    (15*sin(4*x))/256 - (15*sin(6*x))/1024 + (5*sin(8*x))/2048 - ...
    sin(10*x)/5120

```

Problem 7

$$\iint_D (2x^2 \cos^2(xy) + 3x^2 y^2) dx dy = \frac{1135}{8} - \frac{\cos(18)}{8}$$

$$D = \{(x, y) : y \geq 0, x \leq 3, y \geq x\}$$

```

1 %-----Problem 7-----
2 clc; clear; close all;
3
4 % If No Analytical Solution Exists
5 % f = @(x, y) 2 * x.^2 .* (cos(x .* y)).^2 + 3 * x.^2 .* y.^2;
6 % xmin = 0; xmax = 3;
7 % ymin = 0; ymax = @(x) x;
8 % result = integral2(f, xmin, xmax, ymin, ymax);
9
10 syms x y;
11 f = 2 * x^2 * cos(x * y)^2 + 3 * x^2 * y^2;
12 result = int(int(f, y, 0, x), x, 0, 3);
13
14 fprintf('The answer of problem 7 is: ');
15 disp(result);
16
17 % Output:
18 % The answer of problem 7 is: 1135/8 - cos(18)/8

```

Problem 8

$$y(x) = e^{x^2} \cos^2 x \sin(x^2), \quad x = 0$$

$$\begin{aligned}
y(x) &= x^2 + O(x^6) \\
&= -\frac{x^6}{3} + \frac{x^2}{2} + O(x^8) \\
&= -\frac{2x^8}{45} - \frac{x^6}{3} + \frac{x^2}{2} + O(x^{10})
\end{aligned}$$

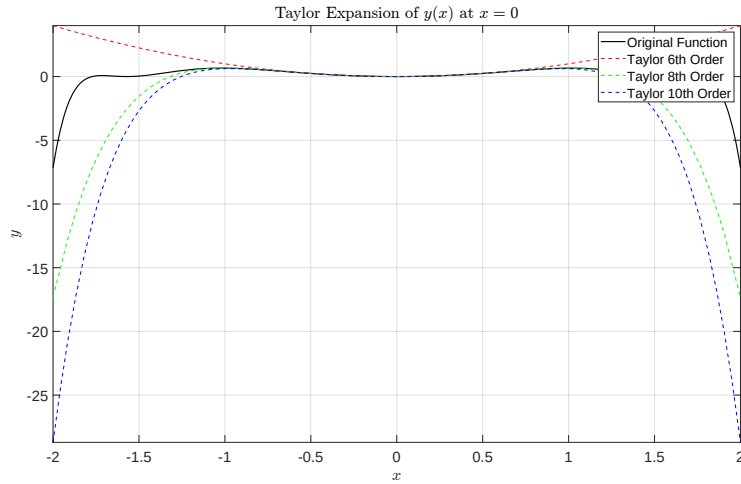


Figure 1: Problem 8

```

1  %-----Problem 8-----
2  clc; clear; close all;
3
4  syms x;
5  y = exp(x^2) * (cos(x))^2 * sin(x^2);
6  % Taylor 6th, 8th, 10th Order
7  T_6 = taylor(y, x, 'Order', 6);
8  T_8 = taylor(y, x, 'Order', 8);
9  T_10 = taylor(y, x, 'Order', 10);
10
11 fprintf('Problem 8, Taylor 6th Order Function: ');
12 disp(T_6);
13 fprintf('Problem 8, Taylor 8th Order Function: ');
14 disp(T_8);
15 fprintf('Problem 8, Taylor 10th Order Function: ');
16 disp(T_10);
17
18 % Plot the Curves
19 fplot(y, [-2,2], 'k', 'LineWidth', 1.4);
20 hold on;
21 fplot(T_6, [-2,2], 'r', 'LineWidth', 1.2);
22 fplot(T_8, [-2,2], 'g', 'LineWidth', 1.2);
23 fplot(T_10, [-2,2], 'b', 'LineWidth', 1.2);

```

```

24 legend('Original Function', 'Taylor 6th Order', 'Taylor 8th ...
    Order', 'Taylor 10th Order');
25 xlabel('$x$', 'Interpreter', 'latex');
26 ylabel('$y$', 'Interpreter', 'latex');
27 title('Taylor Expansion of $y(x)$ at $x=0$', 'Interpreter', ...
    'latex');
28 grid on;
29
30 % Save and Export the Graph
31 set(gca, 'FontSize', 18);
32 set(gcf, 'Position', [100, 100, 1600, 900]);
33 exportgraphics(gcf, 'Problem8.pdf', 'ContentType', 'vector');
34
35 % Output:
36 % Problem 8, Taylor 6th Order Function: x^2
37
38 % Problem 8, Taylor 8th Order Function: - x^6/3 + x^2
39
40 % Problem 8, Taylor 10th Order Function: - (2*x^8)/45 - x^6/3 + x^2

```

Problem 9

$$\begin{cases} y_1' = y_1 - 2y_2 + x \\ y_2' = 3y_1 + y_2 \\ y_1(0) = 1, y_2(0) = 4 \end{cases}$$

$$y_1 = \frac{\sqrt{6}e^x \cos(\sqrt{6}x) \left(\frac{22\sqrt{6}}{49} + \frac{e^{-x} (12 \sin(\sqrt{6}x) + 5\sqrt{6} \cos(\sqrt{6}x) + 42x \sin(\sqrt{6}x) - 7\sqrt{6}x \cos(\sqrt{6}x))}{98} \right)}{3}$$

$$- \frac{\sqrt{6}e^x \sin(\sqrt{6}x) \left(\frac{e^{-x} (12 \cos(\sqrt{6}x) - 5\sqrt{6} \sin(\sqrt{6}x) + 42x \cos(\sqrt{6}x) + 7\sqrt{6}x \sin(\sqrt{6}x))}{98} + \frac{190}{49} \right)}{3}$$

$$y_2 = e^x \sin(\sqrt{6}x) \left(\frac{22\sqrt{6}}{49} + \frac{e^{-x} (12 \sin(\sqrt{6}x) + 5\sqrt{6} \cos(\sqrt{6}x) + 42x \sin(\sqrt{6}x) - 7\sqrt{6}x \cos(\sqrt{6}x))}{98} \right) \\ + e^x \cos(\sqrt{6}x) \left(\frac{e^{-x} (12 \cos(\sqrt{6}x) - 5\sqrt{6} \sin(\sqrt{6}x) + 42x \cos(\sqrt{6}x) + 7\sqrt{6}x \sin(\sqrt{6}x))}{98} + \frac{190}{49} \right)$$

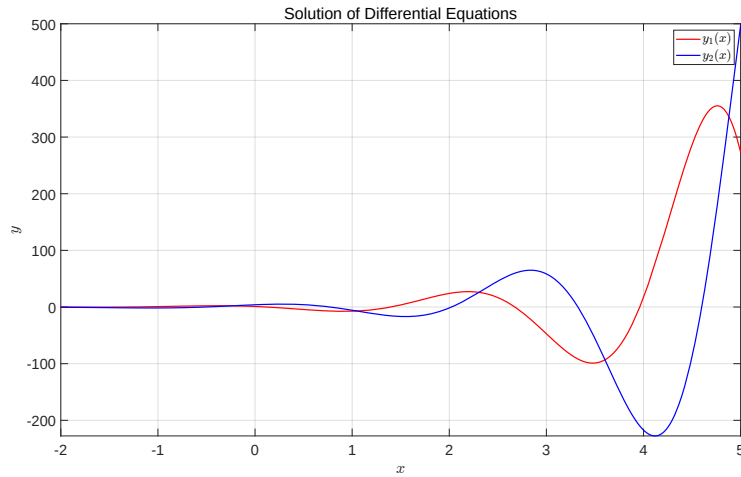


Figure 2: Problem 9

```

1  %-----Problem 9-----
2  clc; clear; close all;
3
4  syms x y1(x) y2(x);
5  eq1 = diff(y1, x) == y1 - 2 * y2 + x;
6  eq2 = diff(y2, x) == 3 * y1 + y2;
7  cond1 = y1(0) == 1;
8  cond2 = y2(0) == 4;
9  sol = dsolve([eq1, eq2], [cond1, cond2]);
10
11 disp('Problem 9, the solution of y1 is:');
12 disp(sol.y1);
13 disp('Problem 9, the solution of y2 is:');
14 disp(sol.y2);
15
16 % Plot the Graph
17 fplot(sol.y1, [-2, 5], 'r', 'LineWidth', 1.5); hold on;
18 fplot(sol.y2, [-2, 5], 'b', 'LineWidth', 1.5);
19 legend('$y_1(x)$', '$y_2(x)$', 'Interpreter', 'latex');
20 xlabel('$x$', 'Interpreter', 'latex');
21 ylabel('$y$', 'Interpreter', 'latex');
22 title('Solution of Differential Equations');
23 grid on;
24
25 % Save and Export the Graph
26 set(gca, 'FontSize', 18);
27 set(gcf, 'Position', [100, 100, 1600, 900]);
28 exportgraphics(gcf, 'Problem9.pdf', 'ContentType', 'vector');
29
30 % Output:
31 % Problem 9, the solution of y1 is:
32 (6^(1/2)*exp(x)*cos(6^(1/2)*x)*((22*6^(1/2))/49 + ...
    (exp(-x)*(12*sin(6^(1/2)*x) + 5*6^(1/2)*cos(6^(1/2)*x) + ...

```

```

33      42*x*sin(6^(1/2)*x) - 7*6^(1/2)*x*cos(6^(1/2)*x))/98))/3 - ...
34      (6^(1/2)*exp(x)*sin(6^(1/2)*x)*((exp(-x)*(12*cos(6^(1/2)*x) - ...
35      5*6^(1/2)*sin(6^(1/2)*x) + 42*x*cos(6^(1/2)*x) + ...
      7*6^(1/2)*x*sin(6^(1/2)*x))/98 + 190/49))/3
33
34 % Problem 9, the solution of y2 is:
35 exp(x)*sin(6^(1/2)*x)*((22*6^(1/2))/49 + ...
      (exp(-x)*(12*sin(6^(1/2)*x) + 5*6^(1/2)*cos(6^(1/2)*x) + ...
      42*x*sin(6^(1/2)*x) - 7*6^(1/2)*x*cos(6^(1/2)*x))/98) + ...
      exp(x)*cos(6^(1/2)*x)*((exp(-x)*(12*cos(6^(1/2)*x) - ...
      5*6^(1/2)*sin(6^(1/2)*x) + 42*x*cos(6^(1/2)*x) + ...
      7*6^(1/2)*x*sin(6^(1/2)*x))/98 + 190/49)

```

Problem 10

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

```

1  %-----Problem 10-----
2  clc; clear; close all;
3
4  syms x y t;
5  % Given Constant
6  A = [2, 3; 4, 6];
7  B = [5; 8];
8
9  % Compute the Rank
10 rank_A = rank(A);
11 rank_AB = rank([A, B]);
12
13 if rank_A == rank_AB
14     if rank_A == size(A, 2)
15         % Unique Solution
16         solution = A \ B;
17         x = rats(solution(1));
18         y = rats(solution(2));
19
20         fprintf('Unique solution: x = %s, y = %s\n', x, y);
21     else
22         % Infinitely Many Solutions
23         P = pinv(A) * B; % Particular Solution
24         N = null(A, 'r'); % General Solution
25         sol_general = P + t * N;
26
27         fprintf(['The equation system has infinitely many ...
28                 solutions: ' ...
29                 'x = %s, y = %s\n'], char(sol_general(1)), ...
30                 char(sol_general(2)));
29     end

```

```

30 else
31     % No Solution
32     fprintf('No solution exists. \n');
33 end
34
35 % Output:
36 % No solution exists.

```

Problem 11

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 1 & 2 \\ 3 & 5 & 1 \end{pmatrix}$$

$$(\eta_1, \eta_2, \eta_3) = \begin{pmatrix} 0.5006 + 0.0000i & -0.3140 - 0.4287i & -0.3140 + 0.4287i \\ 0.5401 + 0.0000i & -0.2343 + 0.4469i & -0.2343 - 0.4469i \\ 0.6765 + 0.0000i & 0.6805 + 0.0000i & 0.6805 + 0.0000i \end{pmatrix}$$

$$(\lambda_1, \lambda_2, \lambda_3) = (7.2121, -2.1061 + 1.3938i, -2.1061 - 1.3938i)$$

$$\det(A) = 46$$

```

1  %-----Problem 11-----
2  clc; clear; close all;
3
4  A = [1, 2, 3; 4, 1, 2; 3, 5, 1];
5  [V, D] = eig(A);
6  a = det(A);
7
8  fprintf('Problem 11, Eigenvector: ');
9  disp(V);
10 fprintf('Problem 11, Eigenvalue: ');
11 disp(D);
12 fprintf('Problem 11, Determinant: ');
13 disp(a);
14
15 % Output:
16 % Problem 11, Eigenvector:    0.5006 + 0.0000i  -0.3140 - 0.4287i ...
    -0.3140 + 0.4287i
17 %    0.5401 + 0.0000i  -0.2343 + 0.4469i  -0.2343 - 0.4469i
18 %    0.6765 + 0.0000i   0.6805 + 0.0000i   0.6805 + 0.0000i
19
20 % Problem 11, Eigenvalue:    7.2121 + 0.0000i   0.0000 + 0.0000i ...
    0.0000 + 0.0000i
21 %    0.0000 + 0.0000i  -2.1061 + 1.3938i   0.0000 + 0.0000i
22 %    0.0000 + 0.0000i   0.0000 + 0.0000i  -2.1061 - 1.3938i
23
24 % Problem 11, Determinant:    46

```

Problem 12

$$(k+1)p_{k+1}(x) = (2k+1)xp_k(x) - kp_{k-1}(x), \quad p_0(x) = 1, \quad p_1(x) = x, \quad k = 1, 2, \dots, 10$$

$$P_0(x) = 1$$

$$P_1(x) = x$$

$$P_2(x) = \frac{3x^2}{2} - \frac{1}{2}$$

$$P_3(x) = \frac{5x^3}{2} - \frac{3x}{2}$$

$$P_4(x) = \frac{35x^4}{8} - \frac{15x^2}{4} + \frac{3}{8}$$

$$P_5(x) = \frac{15x}{8} + \frac{35x^3}{4} + \frac{63x^5}{8}$$

$$P_6(x) = \frac{105x^2}{16} - \frac{315x^4}{16} + \frac{231x^6}{16} - \frac{5}{16}$$

$$P_7(x) = \frac{315x^3}{16} - \frac{35x}{16} + \frac{693x^5}{16} - \frac{429x^7}{16}$$

$$P_8(x) = \frac{3465x^4}{64} - \frac{315x^2}{32} - \frac{3003x^6}{32} + \frac{6435x^8}{128} + \frac{35}{128}$$

$$P_9(x) = \frac{315x}{128} - \frac{1115x^3}{32} + \frac{9009x^5}{64} - \frac{6435x^7}{32} + \frac{12155x^9}{128}$$

$$P_{10}(x) = \frac{3465x^2}{256} - \frac{15015x^4}{128} + \frac{45045x^6}{128} - \frac{109395x^8}{256} + \frac{46189x^{10}}{256} - \frac{63}{256}$$

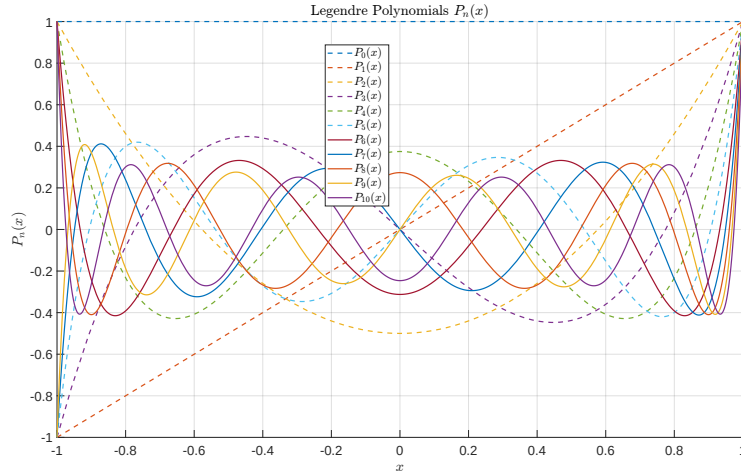


Figure 3: Problem 12

Zero Points:

$$P_1(x) = 0 \quad (0)$$

$$P_2(x) = 0 \quad (\pm \frac{\sqrt{3}}{3})$$

$$P_3(x) = 0 \quad (\pm \frac{\sqrt{15}}{5})$$

$$P_4(x) = 0 \quad (\pm \sqrt{\frac{3}{7} \pm \frac{2\sqrt{30}}{2}})$$

$$P_5(x) = 0 \quad (0, \pm \sqrt{\frac{5}{9} \pm \frac{2\sqrt{70}}{63}})$$

$$P_6(x) = 0 \quad (\pm 0.2386, \pm 0.6612, \pm 0.9325)$$

$$P_7(x) = 0 \quad (0, \pm 0.4058, \pm 0.7415, \pm 0.9491)$$

$$P_8(x) = 0 \quad (\pm 0.1834, \pm 0.5255, \pm 0.7967, \pm 0.9603)$$

$$P_9(x) = 0 \quad (0, \pm 0.3243, \pm 0.6134, \pm 0.8360, \pm 0.9682)$$

$$P_{10}(x) = 0 \quad (0, \pm 0.1489, \pm 0.4334, \pm 0.6794, \pm 0.8651, \pm 0.9739)$$

```

1  %-----Problem 12-----
2  clear; close all;
3
4  syms x;
5  n_max = 10;
6
7  % Initialize Legendre Polynomials
8  P = sym(zeros(n_max+1, 1));
9  P(1) = 1;
10 P(2) = x;
11
12 % Generate Legendre Polynomials
13 for k = 1 : n_max-1
14     P(k+2) = expand(((2*k+1) * x * P(k+1) - k * P(k)) / (k+1));
15 end
16
17 % 0-10th Legendre Polynomials
18 disp('0-10th Legendre Polynomials:');
19 for k = 0 : n_max
20     fprintf('P_%d(x) = %s\n', k, char(P(k+1)));
21 end
22
23 % Calculate Coefficients and Zero Points
24 for k = 1:n_max
25     fprintf('\nP_%d(x) Coefficients: ', k);
26     disp(coefs(P(k+1), x, 'All'));
27     fprintf('P_%d(x) Zero Points: ', k);
28     disp(solve(P(k+1), x));
29     disp(double(solve(P(k+1), x)));
30 end
31

```

```

32 % Plot Legendre Polynomials
33 figure;
34 hold on;
35 title('Legendre Polynomials  $P_n(x)$ ', 'Interpreter', 'latex');
36 xlabel('$x$', 'Interpreter', 'latex');
37 ylabel('$P_n(x)$', 'Interpreter', 'latex');
38 grid on;
39 legend_list = cell(n_max+1, 1);
40
41 for k = 0 : n_max/2
42     fplot(P(k+1), [-1, 1], 'LineWidth', 1.5, 'LineStyle', '—');
43     legend_list{k+1} = sprintf('$P_{%d}(x)$', k);
44 end
45
46 for k = n_max/2 + 1 : n_max
47     fplot(P(k+1), [-1, 1], 'LineWidth', 1.5, 'LineStyle', '-');
48     legend_list{k+1} = sprintf('$P_{%d}(x)$', k);
49 end
50
51 legend(legend_list, 'Interpreter', 'latex', 'Location', 'best');
52 hold off;
53
54 % Save and Export the Graph
55 set(gca, 'FontSize', 16);
56 set(gcf, 'Position', [100, 100, 1600, 900]);
57 exportgraphics(gcf, 'Problem12.pdf', 'ContentType', 'vector');
58
59 % Detailed Output is Omitted

```

For Detailed Output, Click Here.